

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Bharath P(192511282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Bharath

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Industry Problem: Diabetes Diagnosis and Personalised Treatment Recommendation

Introduction

A healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI and family history to develop an AI-based system for early diabetes diagnosis. The system should classify whether a patient is diabetic or non-diabetic and should also recommend suitable treatment actions such as Diet, Exercise, Medication and Monitor. The first part can be solved using supervised machine learning techniques such as Decision Tree and Logistic Regression, while the treatment recommendation can be formulated as a Markov Decision Process and solved using Q-Learning. The complete solution should focus particularly on reducing false-negative diabetes cases, maintaining model interpretability and ensuring patient safety.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from previously observed examples and uses those patterns to predict unseen cases. For this medical dataset, the problem can be formulated as a binary classification problem.

Target Variable:

Diabetes = 1 → Diabetic

Diabetes = 0 → Non-diabetic

Important Features:

Feature	Justification
Glucose Level	High glucose level is strongly associated with diabetes and is an important predictor.
BMI	Higher BMI can be associated with increased diabetes risk.
Age	Diabetes risk can vary with age and therefore age is useful for classification.
Blood Pressure	Provides additional information about the patient's health condition.
Cholesterol	Provides information about metabolic and cardiovascular health.
Family History	A family history of diabetes can indicate increased genetic risk.

The three most important initial features would generally be glucose level, BMI and age, although the final importance should be determined from the trained model.

Inductive Learning Process:

Collect patient records.

Clean and preprocess the data.

Separate input features and the diabetes target.

Divide the data into training and testing sets.

Train a classification model.

Validate the model using appropriate performance measures.

Use the trained model to predict diabetes for new patients.

Thus, the model learns a general relationship between patient characteristics and diabetes status from the available labelled examples.

(b) Handling Class Imbalance

The dataset may contain significantly more non-diabetic patients than diabetic patients. This creates a class imbalance problem. A model trained directly on such data may become biased toward the majority class and predict most patients as non-diabetic.

Suitable Strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) is a suitable approach. SMOTE creates synthetic examples of the minority diabetic class based on existing minority-class observations.

Procedure:

Identify the minority diabetic samples.

Select neighbouring diabetic samples.

Generate synthetic samples between existing minority observations.

Increase the representation of the diabetic class.

Train the model using the balanced training data.

SMOTE is useful because the main clinical concern is detecting diabetic patients correctly. Missing a diabetic patient is a false negative, which can be more serious than incorrectly flagging a non-diabetic patient. SMOTE must be applied only to the training data, after the train-test split, to avoid data leakage. Class weights can also be used as an alternative. Under-sampling is less desirable if it discards useful medical information from the majority class.

(c) 70:30 Train-Test Split and Preprocessing

The dataset can be divided as follows:

70% → Training data

30% → Testing data

The 70:30 split provides enough data for the model to learn patterns while keeping a sufficiently large independent test set for evaluating performance. For a medical application, the test set should not be used

during training because it must represent unseen patients. A stratified split should be used so that the diabetic/non-diabetic ratio is approximately preserved in both training and testing sets.

Preprocessing Steps:

1. Missing Value Handling: Missing numerical values can be handled using median imputation, while categorical missing values can be replaced using the most frequent category.
2. Normalisation: Numerical features such as age, blood pressure, cholesterol, glucose and BMI can be normalised when using models sensitive to feature scale.

$$z = (x - \mu) / \sigma$$

where x is the original value, μ is the mean and σ is the standard deviation.

3. Encoding: Family history can be encoded as No = 0 and Yes = 1. The target can similarly be represented as Non-diabetic = 0 and Diabetic = 1.

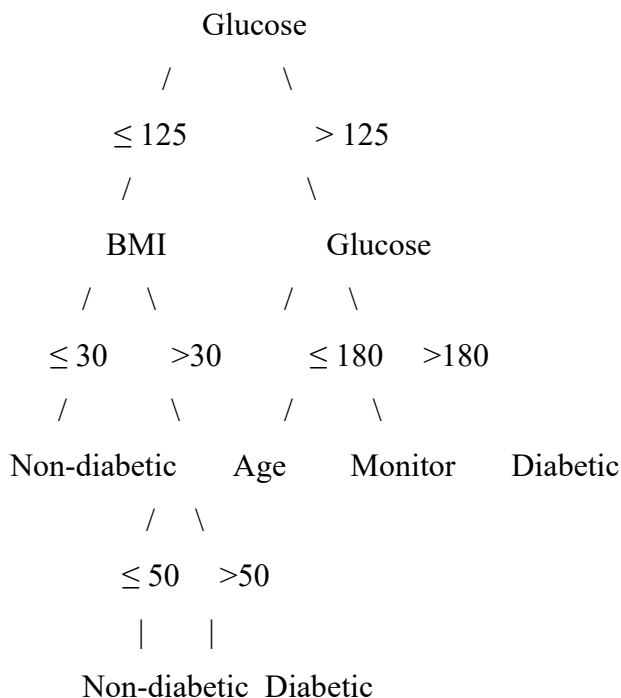
Impact: Preprocessing improves data quality, makes the input suitable for machine learning and prevents numerical differences between features from unnecessarily affecting scale-sensitive algorithms.

TASK 2: DECISION TREE FOR DIAGNOSIS

(a) Decision Tree Classifier

A Decision Tree classifies patients by repeatedly dividing the dataset according to feature values. For the diabetes problem, the tree can use glucose, BMI, age, blood pressure, cholesterol and family history.

Illustrative first three levels of the tree:



The exact tree depends on the actual patient dataset.

Root Node Selection:

The root node should be selected using the feature that produces the largest reduction in impurity. One method is Information Gain.

$$IG(S,A) = H(S) - \sum (|S_v|/|S|) H(S_v)$$

$$H(S) = -\sum p_i \log_2(p_i)$$

Another method is the Gini Index:

$$Gini(S) = 1 - \sum p_i^2$$

The feature with the highest Information Gain or lowest resulting Gini impurity is selected as the root.

Feature	Illustrative Information Gain
Glucose	0.31
BMI	0.18
Age	0.12
Blood Pressure	0.08
Cholesterol	0.06
Family History	0.10

Since glucose has the highest illustrative Information Gain, $IG(\text{Glucose}) = 0.31$, glucose is selected as the root node. These values are illustrative because the assessment file does not provide a numerical patient dataset.

(b) Model Evaluation

The Decision Tree should be evaluated using Accuracy, Precision, Recall and F1-Score.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Illustrative Confusion Matrix:

	Actual Diabetic	Actual Non-diabetic
Predicted Diabetic	TP = 25	FP = 5
Predicted Non-diabetic	FN = 3	TN = 67

$$\text{Total} = 25 + 5 + 3 + 67 = 100$$

$$\text{Accuracy} = (25 + 67) / 100 = 0.92 = 92\%$$

$$\text{Precision} = 25 / (25 + 5) = 0.8333 = 83.33\%$$

$$\text{Recall} = 25 / (25 + 3) = 0.8929 = 89.29\%$$

$$F1 = 2 \times (0.8333 \times 0.8929) / (0.8333 + 0.8929) \approx 0.862 = 86.2\%$$

False Negatives = 3. These are patients who actually have diabetes but are incorrectly classified as non-diabetic.

False negatives can be reduced by using class weights, applying SMOTE to the training set, adjusting the classification threshold, optimising the model for recall, collecting more representative diabetic samples and using a second clinical test for high-risk cases. Therefore, the system should not rely only on accuracy; recall and false-negative rate should receive strong consideration.

(c) Pre-Pruning vs Post-Pruning

A Decision Tree can become too complex and overfit the training data.

Pre-pruning limits tree growth during training using maximum tree depth, minimum samples required for splitting and minimum samples per leaf. It provides faster training and a smaller tree, but important patterns may be removed too early.

Post-pruning first builds a larger tree and then removes branches that provide insufficient predictive benefit. One common method is cost-complexity pruning.

$$R_{\alpha}(T) = R(T) + \alpha|T|$$

where $R(T)$ is tree error, $|T|$ is the number of terminal nodes and α is the complexity parameter.

Model	Illustrative Accuracy	Complexity
Fully grown Tree	94% training / 84% testing	High
Pre-pruned Tree	90%	Medium
Post-pruned Tree	92%	Low-Medium

The fully grown tree may achieve high training accuracy but lower testing accuracy because of overfitting. A post-pruned Decision Tree is preferable for deployment when it maintains good predictive performance while reducing unnecessary complexity and remains easier for healthcare professionals to interpret.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Logistic Regression

Logistic Regression can be applied to the same diabetes dataset as a statistical learning model. The probability of diabetes is:

$$P(Y = 1) = 1 / (1 + e^{(-z)})$$

$$\text{where } z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n.$$

For this problem, $z = \beta_0 + \beta_1(\text{Glucose}) + \beta_2(\text{BMI}) + \beta_3(\text{Age}) + \beta_4(\text{BP}) + \beta_5(\text{Cholesterol}) + \beta_6(\text{FamilyHistory})$.

For example, if $P(\text{Diabetes}) = 0.82$ and the selected threshold is 0.5, then $0.82 > 0.5$ and the patient is classified as diabetic.

Model	Illustrative Accuracy	Illustrative AUC-ROC
Decision Tree	92%	0.90

The values above are illustrative because the assessment file does not provide a numerical patient dataset from which actual model results can be calculated.

AUC-ROC represents the model's ability to distinguish between diabetic and non-diabetic patients. An AUC closer to 1 indicates excellent discrimination, while $AUC = 0.5$ indicates performance close to random classification.

Logistic Regression may be preferable when probability estimates are important, interpretability is required, the effect of individual features needs to be studied, and a simpler model is preferred for clinical risk scoring. Decision Trees are preferable when non-linear relationships and easy rule-based interpretation are important.

(b) Feature Importance Analysis

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The models agree that glucose and BMI are among the strongest predictors. The third predictor may differ because the two models measure relationships differently. A Decision Tree measures reduction in impurity produced by a feature during splitting, whereas Logistic Regression estimates the relationship between predictors and the log-odds of the outcome. Therefore, differences in the top-three rankings are possible.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) Modelling Treatment Recommendation as an MDP

The treatment recommendation problem can be represented using a Markov Decision Process (MDP). An MDP consists of (S, A, R, P, γ) , where S represents states, A represents actions, R represents rewards, P represents transition probabilities and γ represents the discount factor.

1. States:

State	Description
S1	Low-risk / controlled
S2	Moderate-risk
S3	High-risk
S4	Poorly controlled
S5	Improved / stable

2. Actions: $A = \{\text{Diet, Exercise, Medication, Monitor}\}$. These are the actions specified in the assessment.
3. Reward Function: The reward should represent improvement in patient health.

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10

4. Transition Dynamics: Transition dynamics describe how the patient moves from one state to another after an action. $P(S_{t+1} | S_t, A_t)$ represents the probability of reaching the next state given the current state and selected action.

Example: Moderate Risk + Diet $\rightarrow$ Improved / Stable. High Risk + Monitor $\rightarrow$ High Risk. In a real system, transition probabilities must be estimated from reliable longitudinal clinical data.

(b) Q-Learning

Q-Learning is a model-free reinforcement learning algorithm that learns the expected future reward for each state-action pair.

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where $Q(s,a)$ is the current Q-value, α is the learning rate, r is the reward, γ is the discount factor, s' is the next state and $\max Q(s',a')$ is the best future Q-value.

Assume $\alpha = 0.5$, $\gamma = 0.9$ and initially all Q-values are 0.

Five illustrative patient episodes:

Episode	Initial State	Action	Next State	Reward
1	Moderate Risk	Diet	Improved	+10
2	Moderate Risk	Exercise	Improved	+8
3	High Risk	Medication	Moderate Risk	+6
4	Moderate Risk	Monitor	Moderate Risk	0
5	High Risk	Exercise	High Risk	-2

These episodes are an illustrative demonstration of the required Q-learning procedure and are not clinical recommendations.

Q-Table Update – Iteration 1

1. Moderate Risk + Diet: $Q_{\text{new}} = 0 + 0.5[10 + 0.9(0) - 0] = 5$. Therefore $Q(\text{Moderate, Diet}) = 5$.
2. Moderate Risk + Exercise: $Q_{\text{new}} = 0 + 0.5[8 + 0.9(0) - 0] = 4$. Therefore $Q(\text{Moderate, Exercise}) = 4$.
3. High Risk + Medication: $Q_{\text{new}} = 0 + 0.5[6 + 0.9(0) - 0] = 3$. Therefore $Q(\text{High, Medication}) = 3$.

State	Diet	Exercise	Medication	Monitor
Moderate Risk	5.0	4.0	0	0
High Risk	0	0	3.0	0

Q-Table Update – Iteration 2

1. Moderate Risk + Exercise: Current $Q = 4$, $r = 8$, and max next-state $Q = 5$.

$$Q_{\text{new}} = 4 + 0.5[8 + 0.9(5) - 4] = 4 + 0.5(8.5) = 8.25.$$

2. High Risk + Medication: Current $Q = 3$, $r = 6$, and max next-state $Q = 5$.

$$Q_{\text{new}} = 3 + 0.5[6 + 0.9(5) - 3] = 3 + 0.5(7.5) = 6.75.$$

3. Moderate Risk + Diet: Current $Q = 5$, $r = 10$, and max next-state $Q = 0$.

$$Q_{\text{new}} = 5 + 0.5[10 + 0.9(0) - 5] = 7.50.$$

State	Diet	Exercise	Medication	Monitor
Moderate Risk	7.50	8.25	0	0
High Risk	0	0	6.75	0

Convergence Criterion:

The Q-learning process can be considered approximately converged when the maximum change in Q-values between successive iterations becomes smaller than a predefined threshold, for example $\max |Q_{\text{new}} - Q_{\text{old}}| < 0.01$ for a sufficiently large number of consecutive updates. Another practical criterion is that the learned policy remains unchanged over several episodes.

(c) Final Learned Policy

The policy is obtained by selecting the action with the maximum Q-value for each state:

$$\pi(s) = \arg \max_a Q(s,a)$$

For Moderate Risk: $Q(\text{Diet}) = 7.50$ and $Q(\text{Exercise}) = 8.25$. Therefore $\pi(\text{Moderate Risk}) = \text{Exercise}$.

For High Risk: $Q(\text{Medication}) = 6.75$. Therefore $\pi(\text{High Risk}) = \text{Medication}$.

Patient State	Illustrative Learned Action
Moderate Risk	Exercise
High Risk	Medication

This is only an example of how a policy is extracted from a Q-table. It must not be interpreted as a real medical treatment recommendation. In an actual healthcare system, treatment decisions must be determined and supervised by qualified healthcare professionals.

Ethical Considerations:

Patient Safety: Incorrect recommendations can cause harm. The AI system should not independently make unsafe clinical decisions.

Bias: If the training data is biased or does not adequately represent different patient groups, the model may produce unfair recommendations.

Explainability: Healthcare professionals should understand why the system selected a particular recommendation.

Privacy: Patient health records contain sensitive information, so security, access control and privacy protection are required.

Human Oversight: The RL agent should operate as a decision-support system, not as an uncontrolled autonomous treatment system.

Validation: The model must be tested extensively before deployment for safety, reliability, bias and performance.

Overall Conclusion

The proposed AI system combines supervised machine learning and reinforcement learning to support diabetes diagnosis and treatment decision-making. Inductive learning can use patient features such as glucose level, BMI, age, blood pressure, cholesterol and family history to predict diabetes. A 70:30 stratified train-test split, appropriate preprocessing and class-balancing techniques such as SMOTE can improve model development. A Decision Tree provides an interpretable classification model, while Logistic Regression provides a statistical alternative with useful probability and risk estimates.

For medical diagnosis, accuracy alone is insufficient because false-negative diabetes cases can be clinically important. Therefore, recall, precision, F1-score and AUC-ROC should all be considered. Post-pruning can reduce Decision Tree complexity and improve generalisation.

For treatment recommendation, the problem can be represented as an MDP containing patient states, treatment actions, rewards and transition dynamics. Q-Learning can then learn the expected value of different actions through repeated episodes. The final policy is obtained by selecting the action with the highest Q-value.

However, an AI system for healthcare must not replace qualified medical professionals. Patient safety, fairness, privacy, explainability and human supervision are essential before any such system is deployed. The final system should therefore be designed as a validated clinical decision-support tool, with medical professionals retaining responsibility for patient care.

Note on Numerical Results

The assessment document specifies the required tasks and methods but does not include an actual numerical patient dataset. Therefore, the Information Gain values, confusion-matrix values, model comparison values and Q-learning episodes used above are illustrative examples for demonstrating the required calculations. Actual experimental results should be generated from the supplied patient dataset if one is provided.